

HDOC-16

Question:

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Define **one separate user-defined function for each menu option**, with void return type and with a simple function call (without call by value/reference). Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- Check whether its digits are in increasing order from left to right.
- Check whether its digits are in decreasing order from left to right.
- Find the largest digit and the smallest digit.
- Find the second largest digit and second smallest digit in the number.

Task1: Algorithm:

Step 1: Start.

Step 2: Declare choice as an integer.

Step 3: Display the menu:

1. Check digits in increasing order.
2. Check digits in decreasing order.
3. Find largest and smallest digit.
4. Find second largest and second smallest digit.

Step 4: Input choice.

Step 5: Use switch(choice):

- Case 1: Call increasing().
 - Input the positive integer inside the function.
 - Compare adjacent digits.
 - Display whether the digits are in increasing order from left to right.
- Case 2: Call decreasing().
 - Input the positive integer inside the function.
 - Compare adjacent digits.
 - Display whether the digits are in decreasing order from left to right.
- Case 3: Call largestSmallest().
 - Input the positive integer.
 - Extract each digit using % 10.
 - Compare each digit with largest and smallest.
 - Display the largest and smallest digits.

- Case 4: Call secondLargestSmallest().
 - Input the positive integer.
 - Extract every digit.
 - Find the largest, second largest, smallest, and second smallest distinct digits.
 - Display the second largest and second smallest digits.
- Default: Display "Invalid choice".

Step 6: Stop.

Task2: Test Case Table:

Test Case	Choice	Number	Expected Output	Actual Output	Result
1	1	12345	Digits are in increasing order	Digits are in increasing order	Pass
2	2	97531	Digits are in decreasing order	Digits are in decreasing order	Pass
3	3	41673	Largest = 7, Smallest = 1	Largest = 7, Smallest = 1	Pass
4	4	7777	Second largest and second smallest do not exist	Second largest and second smallest do not exist	Pass
5	5	—	Invalid choice	Invalid choice	Pass

“Task3: C Program:

```
#include <stdio.h>
/* Function declarations */
void increasing();
void decreasing();
void largestSmallest();
void secondLargestSmallest();
int main()
{
    int choice;
    printf("MENU\n");
    printf("1. Check increasing order\n");
    printf("2. Check decreasing order\n");
```

```

printf("3. Find largest and smallest digit\n");
printf("4. Find second largest and second smallest digit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch(choice)
{
    case 1:
        increasing();
        break;
    case 2:
        decreasing();
        break;
    case 3:
        largestSmallest();
        break;
    case 4:
        secondLargestSmallest();
        break;
    default:
        printf("Invalid choice\n");
}
return 0;
}
/* Function to check increasing order */
void increasing()
{
    int num, temp, digit, previous;
    int flag = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    temp = num;

    previous = temp % 10;
    temp = temp / 10;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit >= previous)
        {
            flag = 0;
            break;
        }
        previous = digit;
        temp = temp / 10;
    }
}

```

```

    if(flag == 1)
        printf("Digits are in increasing order\n");
    else
        printf("Digits are not in increasing order\n");
}

/* Function to check decreasing order */
void decreasing()
{
    int num, temp, digit, previous;
    int flag = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    temp = num;
    previous = temp % 10;
    temp = temp / 10;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit <= previous)
        {
            flag = 0;
            break;
        }
        previous = digit;
        temp = temp / 10;
    }
    if(flag == 1)
        printf("Digits are in decreasing order\n");
    else
        printf("Digits are not in decreasing order\n");
}

/* Function to find largest and smallest digit */
void largestSmallest()
{
    int num, temp, digit;
    int largest = 0;
    int smallest = 9;

    printf("Enter a positive integer: ");
    scanf("%d", &num);
    temp = num;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit > largest)

```

```

        largest = digit;
        if(digit < smallest)
            smallest = digit;

        temp = temp / 10;
    }
    printf("Largest digit = %d\n", largest);
    printf("Smallest digit = %d\n", smallest);
}

/* Function to find second largest and second smallest digit */
void secondLargestSmallest()
{
    int num, temp, digit;
    int largest = -1;
    int secondLargest = -1;
    int smallest = 10;
    int secondSmallest = 10;

    printf("Enter a positive integer: ");
    scanf("%d", &num);
    temp = num;
    while(temp > 0)
    {
        digit = temp % 10;
        /* Find largest and second largest */
        if(digit > largest)
        {
            secondLargest = largest;
            largest = digit;
        }
        else if(digit > secondLargest && digit != largest)
        {
            secondLargest = digit;
        }
        /* Find smallest and second smallest */
        if(digit < smallest)
        {
            secondSmallest = smallest;
            smallest = digit;
        }
        else if(digit < secondSmallest && digit != smallest)
        {
            secondSmallest = digit;
        }
    }
}

```

```

        temp = temp / 10;
    }

    if(secondLargest == -1 || secondSmallest == 10)
    {
        printf("Second largest and second smallest digits do not exist\n");
    }
    else
    {
        printf("Second largest digit = %d\n", secondLargest);
        printf("Second smallest digit = %d\n", secondSmallest);
    }
}

```

Task4: Output:

```

(kali@kali)-[~/hdoc16.c]
$ ./a.out
MENU
1. Check increasing order
2. Check decreasing order
3. Find the largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 1
Enter a positive integer: 12345
Digits are in increasing order

```

```

(kali@kali)-[~/hdoc16.c]
$ ./a.out
MENU
1. Check increasing order
2. Check decreasing order
3. Find the largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 2
Enter a positive integer: 97531
Digits are in decreasing order

```

```
(kali㉿kali)-[~/hdoc16.c]
$ ./a.out
MENU
1. Check increasing order
2. Check decreasing order
3. Find the largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 3
Enter a positive integer: 41673
Largest digit = 7
Smallest digit = 1
```

```
(kali㉿kali)-[~/hdoc16.c]
$ ./a.out
MENU
1. Check increasing order
2. Check decreasing order
3. Find the largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 4
Enter a positive interger: 7777
Second largest and second smallest digits do not exist
```

```
(kali㉿kali)-[~/hdoc16.c]
$ ./a.out
MENU
1. Check increasing order
2. Check decreasing order
3. Find the largest and smallest digit
4. Find second largest and second smallest digit
Enter your choice: 5
Invalid choice
```